

When the Signal Survives but the Meaning Does Not

Information Theory, Compression, and Semantic Fidelity

Reality Drift Framework — Bridge Brief #5

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The Established Insight

Every communication system faces a basic problem.

Information must survive movement.

A message is encoded.

The signal passes through a channel.

Noise enters.

The message is decoded.

The system succeeds when enough information survives for the message to be recovered.

source → encoding → transmission → decoding → recovered signal

Information theory provided a rigorous framework for understanding this process.

Its central concern is signal preservation.

How much information can be transmitted?

How much noise can the channel tolerate?

What can be compressed?

What must be preserved for the signal to remain recoverable?

This framework made modern telecommunications, computing, networking, and digital storage possible.

What Information Theory Explains

Information theory explains how signals survive transformation.

It provides a language for:

- transmission
- noise
- encoding
- decoding
- compression
- information loss
- error correction
- channel capacity

The underlying problem is structural.

A message enters a system in one form and emerges in another. The system must preserve enough of the original information for the output to remain usable.

The question is:

Was the signal preserved?

This is a problem of fidelity.

But it is not yet a problem of meaning.

The Boundary

A signal can survive perfectly while its meaning changes.

A sentence may be copied without error.

A document may remain unchanged.

A metric may be transmitted accurately.

A model may reproduce the same output consistently.

None of this guarantees that the representation still means the same thing in context.

The words may survive while their reference changes.

The number may survive while the condition it once tracked changes.

The narrative may remain coherent while becoming detached from the events that originally supported it.

The file remains intact.

Its grounding weakens.

This is the boundary between **signal fidelity** and **Semantic Fidelity**.

The Reality Drift Extension

Reality Drift asks a different question from information theory.

Information theory asks:

Was the information preserved?

Reality Drift asks:

Was the relationship between the representation and reality preserved?

This distinction matters because meaning is not contained only in the signal.

Meaning also depends on:

- context
- reference
- intention
- assumptions
- relationships among concepts
- conditions outside the representation

A message can survive transmission while those surrounding structures change.

The representation remains available.

Its interpretation shifts.

The information survives.

The meaning drifts.

Compression

Compression makes this problem more visible.

All representations compress reality.

A map removes detail.

A metric reduces a complex outcome to a number.

A summary reduces an argument to a smaller set of claims.

A dashboard reduces an organization to visible indicators.

A model reduces a changing environment to selected variables and relationships.

Compression is unavoidable.

No representation can preserve everything.

The important question is not whether something was removed.

It is whether the removed material was necessary for preserving meaning.

As compression increases:

distinctions disappear → assumptions become implicit → interpretation carries more weight

The signal may remain clean.

The representation may become more efficient.

The relationship to reality becomes harder to evaluate.

Semantic Fidelity

Semantic Fidelity is the degree to which meaning survives transformation.

It asks whether a representation continues preserving:

- the original intent
- the relevant context
- important distinctions
- uncertainty
- conceptual relationships
- the connection between the representation and what it refers to

Signal fidelity is concerned with whether the representation survived.

Semantic Fidelity is concerned with whether the representation still carries the meaning required for reliable judgment.

The two can diverge.

perfect transmission + weakened reference = preserved signal, degraded meaning

The Bridge

Information theory and Reality Drift address different layers of transformation.

Information theory asks:

How can information survive transmission through a noisy channel?

Reality Drift asks:

How can meaning survive repeated compression, interpretation, optimization, and reuse?

Information theory explains why a message can be recovered.

Reality Drift examines whether the recovered message still refers reliably to the same underlying reality.

This becomes especially important in systems where representations are repeatedly transformed:

source material → summary → classification → model output → report → decision

Each stage may preserve recognizable information.

Each stage may also alter emphasis, remove context, introduce assumptions, or change what the representation is used to mean.

Where the Distinction Matters

Artificial Intelligence

An AI system generates a fluent and internally structured answer.

The language is intact.

The relationship between the answer and the underlying evidence remains uncertain.

The signal is coherent.

The grounding is weak.

Summarization

A document is compressed into a shorter representation.

The main claims remain visible.

The assumptions, uncertainty, and argumentative structure disappear.

Information survives.

Meaning narrows.

Institutional Reporting

Data is transmitted accurately into dashboards and reports.

The numbers are correct.

The relationship between the reported indicators and lived conditions becomes harder to verify.

Signal fidelity remains high.

Semantic Fidelity weakens.

Narrative Drift

A story is repeated consistently across time.

The same language, categories, and structure survive.

The conditions that originally gave the narrative meaning have changed.

The narrative remains stable.

Its reference has drifted.

Core Distinction

Information theory explains how signals survive transmission.

Semantic Fidelity asks whether meaning survives transformation.

Reality Drift begins when representations remain coherent, accurate, and recoverable while their relationship to the realities they describe becomes progressively less certain.